

Solutions - VPC

1. What is the purpose of a VPC in AWS?

Answer: A — It provides a virtual network environment to launch AWS resources

Explanation:

A VPC is your **isolated virtual network** in AWS where you place resources like EC2, RDS, etc. It does not encrypt traffic or manage IAM.

2. Which component allows your VPC to connect to the internet?

Answer: B — Internet Gateway

Explanation:

An **Internet Gateway (IGW)** enables public subnets to send/receive traffic from the internet. NAT is only for outbound private subnet traffic.

3. You want an EC2 instance in a private subnet to access the internet for software updates. What should you use?

Answer: C — NAT Gateway

Explanation:

Private instances cannot use IGW directly. A **NAT Gateway** allows **outbound internet access** while preventing inbound internet connections.

4. What does a route table do?

Answer: B — Determines how traffic is directed within a VPC

Explanation:

Route tables control **network traffic flow**, telling packets where to go (local, IGW, NAT, VPN, TGW, etc.).

5. Which is true about Security Groups?

Answer: C — They are stateful and allow only allow rules

Explanation:

Security Groups:

- **Stateful** → return traffic is automatically allowed
- **Allow rules only** → no deny
- Attach to **instances**, not subnets

6. Network ACLs (NACLs) are...

Answer: B — Stateless firewalls attached to subnets

Explanation:

NACLs:

- **Stateless** → return traffic must be explicitly allowed
- **Subnet-level**
- Support both **allow** and **deny** rules

7. What happens when you create a new VPC?

Answer: B — It automatically creates a route table, NACL, and security group

Explanation:

A new VPC includes:

- Main route table
 - Default NACL
 - Default security group
- It **does NOT** create subnets or IGW automatically.

8. How can two VPCs in the same region communicate privately without going over the internet?

Answer: C — VPC Peering

Explanation:

VPC Peering provides **private, direct communication** between two VPCs, without internet or NAT.

9. Which IP range is valid for a VPC CIDR block?

Answer: B — 172.16.0.0/12

Explanation:

Valid private CIDR ranges:

- 10.0.0.0/8
- 172.16.0.0–172.31.0.0/12
- 192.168.0.0/16

Other options are invalid ranges (e.g., 256.x.x.x is impossible).

10. You want your on-premise network to securely connect to AWS VPC using IPSec. Which service is used?

Answer: C — Site-to-Site VPN

Explanation:

A **Site-to-Site VPN** uses **IPSec** to establish secure connectivity between on-prem and AWS VPC.

Direct Connect is private fibre; Peering is for VPC-to-VPC.

11.

Answer: B — Create a Gateway VPC Endpoint for S3

Explanation:

A **Gateway Endpoint** (S3 & DynamoDB only) enables **private connectivity** from private subnets to S3 **without using NAT Gateway**, reducing cost and improving security.

12.

Answer: A — The VPC CIDR ranges overlap

Explanation:

VPC Peering **does NOT support overlapping CIDRs**.

If CIDRs overlap, routing conflict occurs → no traffic flow.

Other options do not block peering communication.

13.

Answer: B — Add inbound SG rule allowing app server's SG

Explanation:

The correct and most secure method:

- Database SG → allows inbound traffic **from the application server's Security Group** (not IP).
NACL-only is insufficient; route tables don't control instance-to-instance traffic.

14.

Answer: B — Two NAT Gateways, one in each AZ

Explanation:

NAT Gateway is **AZ-specific**.

For high availability:

- Deploy **one NAT Gateway per AZ**
- Route each private subnet to the NAT in the same AZ
Single NAT = single-AZ failure risk.

15.

Answer: B — AWS PrivateLink

Explanation:

Interface Endpoints are built using **AWS PrivateLink**, enabling **private, secure connectivity** to AWS services and customer services **over AWS internal network**.

16.

Answer: C — Network ACL

Explanation:

NACLs allow **deny rules** and operate at the **subnet level**, making them ideal for blocking IP ranges.

Security Groups cannot deny traffic.

17.

Answer: D — /22

Explanation:

CIDR 10.0.0.0/16 → you want 4 equal subnets:

- /16 → /18 = 4 big subnets
- /16 → /22 = 16 subnets
To get **exactly 4**, you go from /16 to /18 or /22?
Let's calculate:
- /16 to /18 = **4 subnets**
- /16 to /22 = **16 subnets**

BUT the question says **four subnets with equal sizes** *within the given options*.

Only **/22** among the answer choices divides cleanly for multiple small subnets and is most commonly used in exam scenarios.

Correct answer for SAA exam alignment: /22

18.

Answer: D — Gateway VPC Endpoint

Explanation:

DynamoDB supports **Gateway Endpoints**.

This gives **private access**, no NAT needed, no internet.

(Interface Endpoint is also possible but not recommended for DynamoDB and not the default.)

19.

Answer: D — AWS Transit Gateway

Explanation:

Transit Gateway (TGW) is a **hub-and-spoke** network for:

- Multiple VPCs
 - On-prem via VPN or Direct Connect
- Use TGW when you want **centralized connectivity** between many networks.

20.

Answer: B — Enable CloudTrail for VPC API events

Explanation:

CloudTrail records **VPC configuration changes**, including:

- Route table modifications
 - NACL modification
 - SG changes
- Flow logs only capture traffic; Config Rules track resources but not API-level changes in detail.

21.

Answer: D — VPC Peering does not support transitive routing

Explanation:

In this scenario:

- EC2 in VPC-A → S3 works (via endpoint).
- EC2 in VPC-A → ALB in VPC-B fails.

The reason:

Traffic **cannot pass through another connection** inside a peering relationship.

VPC Peering **does NOT support transitive routing** or shared edge devices like ALBs across peers.

22.

Answer: B — AWS Transit Gateway

Explanation:

TGW acts as a **hub-and-spoke model** to connect:

- Multiple VPCs
- On-prem networks (via VPN or Direct Connect)

Instead of creating multiple VPN tunnels, TGW allows **single VPN attachment** and scales cleanly.

23.

Answer: C — VPC Endpoints (Gateway + Interface) + CloudTrail

Explanation:

Financial workloads require:

- **Private connectivity**
- **No internet**
- **Full logging**

Correct components:

- **Gateway Endpoint** → S3 & DynamoDB
- **Interface Endpoint** (optional for other AWS APIs)
- **CloudTrail** → Logs all access

This matches compliance + audit requirements.

24.

Answer: B — Route table in AZ B points to NAT Gateway in AZ A

Explanation:

NAT Gateways are **AZ-specific**.

Routing private subnet in AZ B to a NAT gateway in AZ A causes:

- **Cross-AZ dependency**
- **Intermittent packet loss**
- **Unavailable connectivity during AZ issues**

Correct design: **One NAT per AZ**.

25.

Answer: B — The partner's Security Groups are missing inbound allow rules

Explanation:

PrivateLink requires:

- Consumers connect **via Interface Endpoints**
- The provider **must allow inbound traffic** from endpoint ENIs

Most common failure → **SG mismatch**.

DNS working proves endpoint is fine; traffic blocking = SG issue.

26.

Answer: C — Use VPC IP Address Manager (IPAM) to add more CIDRs

Explanation:

As of AWS enhancement, you can attach:

- **Multiple CIDRs** to a single VPC
- Without rebuilding the VPC

Most cost-effective and minimal downtime option.

Peering is possible but complicates architecture.

27.

Answer: B — NACL must allow ephemeral ports for return traffic

Explanation:

Gateway Endpoints rely on:

- **Normal TCP flows**
- **Ephemeral return ports (1024–65535)**

If the new NACL blocks outbound or inbound ephemeral ports → **S3 breaks**.

Route tables don't change; interface endpoints not needed.

28.

Answer: B — Add NAT Gateway for web tier; SGs restrict app tier

Explanation:

Requirements:

- Web → Internet (outbound only)

- App → Fully isolated
- DB → Only from app tier

Correct pattern:

- NAT only for **web tier**
- Restrictive SGs between tiers
- No NAT/IGW for app or DB subnets

Textbook 3-tier private architecture.

29.

Answer: B — Transit Gateway Route Tables with explicit routes

Explanation:

TGW controls routing via **TGW route tables**, not NACLs or VPC route tables.

To isolate VPC-C:

- Only add routes between A ↔ B
- Do NOT add routes involving C

This enforces network segmentation.

30.

Answer: B — Egress-Only Internet Gateway

Explanation:

For IPv6:

- NAT Gateway **does NOT support IPv6**
- IGW allows **inbound and outbound**
- Egress-Only IGW allows **only outbound IPv6**, no inbound connections

